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# Effect of modified-live porcine reproductive and respiratory syndrome virus (PRRSv) vaccine on the shedding of wild-type virus from an infected population of growing pigs

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ABSTRACT

There are ongoing efforts to eliminate porcine reproductive and respiratory syndrome virus (PRRSv) from regions in the United States swine industry. However, an important challenge for the accomplishment of those efforts is the re-infection of pig units due to the area spread of PRRSv. The objective of this study was to evaluate the effect of PRRS modified-live virus vaccine (MLV) on viral shedding and on dynamics of PRRSv infection in pig populations raised under commercial conditions. The study composed of two rooms of 1000 pigs each. Ten percent of pigs of each room were inoculated with a field isolate of PRRSv. Rooms had separate air spaces and strict scientifically validated biosecurity protocols were adopted to avoid movement of pathogens between rooms. At 8 and 36 dpi (days post inoculation), all pigs of the challenge-vaccine group were inoculated with a MLV vaccine. Pigs of the challenge-control group were placebo-inoculated. Blood and oral fluid samples were collected from each room at 0, 8, 36, 70, 96 and 118 dpi for PRRSv RNA detection using PCR. PRRSv-antibodies were also screened from blood serum samples with a commercially available ELISA test. Additionally, tonsil scraping samples were collected from both groups at 70, 96 and 118 dpi. Moreover, air samples were collected 6 times per week from 0 to 118 dpi and were tested for PRRSv RNA using qPCR assay. There was no difference in the PRRSv infection dynamics measured as duration and magnitude of viremia and seroconversion. Also, there was no difference in the frequency of tonsil scraping samples PRRSv-positive by PCR. However, the challenge-vaccine group had significantly less PRRSv shed compared to the challenge-control group. The challenge-vaccine group had significant less PRRSv-positive oral fluids at 36 dpi. Moreover, the challenge-vaccine group had significant reduction in the cumulative PRRSv shed in the air.

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## 1. Introduction

Porcine reproductive and respiratory syndrome (PRRS) is one the most important swine infectious diseases worldwide. It was first described two decades ago in the United States and Europe [1,2] and has been reported in many pig producing countries [3]. The annual loss associated with PRRS in the United States swine industry was estimated at approximately USD$ 664 million [4]. The etiologic agent, PRRS virus (PRRSv) is a swine specific positive-sense single-stranded enveloped RNA virus belonging to the Arteriviridae family [5].

Upon infection of a susceptible pig, PRRSv replicates extensively in alveolar macrophages [6], resulting in a viremia that lasts approximately 3–5 weeks [7]. Nursery pigs are more susceptible than finishing and adult pigs to develop PRRS clinical signs [8]. However, all ages develop viremia and antibody response in the face of PRRSv infection [9,10]. Following the cessation of viremia, PRRSv establishes a prolonged infection in lymphoid tissues that has been reported up to 6 months in duration [11]. The end of the prolonged infection results in the development of a protective immune response, which clears the infection and confers protection to subsequent challenges [12,13].

PRRSv is shed in different body fluids including oral fluids [14] and bioaerosols [15,16]. It has been reported that PRRSv remains infectious after being transported for up to 9.1 km in the air coming out of finishing barns housing pigs infected with PRRSv [17]. Aerosol transmission is one of the most important PRRSv

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transmission routes because it is unaffected by standard biosecurity protocols employed by farms such as quarantine, change of coveralls and equipment disinfection. Therefore, farms located nearby sites where PRRSv is prevalent are at risk of PRRSv airborne infection [18,19]. Over the past years, a limited but important number of sow farms in the United States were equipped with air filters to prevent airborne PRRSv infection [20]. However, filtering a farm requires substantial investment and growing pig sites are not yet filtered.

To reduce the risk of PRRSv infection of growing pig sites, more information is needed regarding the ability to reduce PRRSv shedding and the resulting viral load into the environment. One potential strategy involves the use of vaccination. Studies have shown that use of modified-live virus (MLV) vaccine prior to PRRSv infection significantly reduces both lung lesions and PRRS clinical signs [21,22]. MLV vaccination also reduces PRRSv shedding in semen [23] and decreases magnitude and duration of viremia of growing pigs [24]. Furthermore, Cano et al. [12,13] demonstrated that PRRS MLV vaccine significantly reduced the duration of shedding in animals previously infected with wild-type PRRSv. However, those experiments were conducted under controlled conditions with a limited number of animals and did not measure aerosol shedding which could represent a risk for neighboring farms.

Therefore, the purpose of this study was to evaluate PRRSv MLV vaccine as a tool to reduce viral shedding as well as duration and magnitude of viremia in infected populations of growing pigs raised under large commercial conditions.

## 2. Design and methods

### 2.1. Animals

Two thousand and fourteen piglets of approximately 3 weeks of age were obtained from two PRRSv-free breeding herds of the same production system consisting each of 6000 C-42 sows (Pig Improvement Company) crossed with PIC 410 boar line (Pig Improvement Company). The source breeding herds were vaccinated with Ingelvac® CircoFLEX (Boehringer Ingelheim Vetmedica Inc.), Ingelvac® MycoFLEX (Boehringer Ingelheim Vetmedica Inc.) and Flusure XP (Pfizer Animal Health). All piglets received at weaning a full dose of Ingelvac® CircoFLEX and Ingelvac® MycoFLEX vaccines. All the procedures with animals described in this study were previously approved by the University of Minnesota Institutional Animal Care and Use Committee (IACUC).

### 2.2. Housing

Pigs were housed in the Pipestone Veterinary Clinic's tunnel ventilated research barn located in Pipestone, Minnesota. The research barn consisted of two rooms (west and east) separated by a solid wall and each room had independent attic and slurry pit. Both rooms had 42 pens of 2.50 m × 7.0 m (17.5 $m^2$), which were divided by metal gates. There were 23–25 pigs per pen. Waterers and feeders of both rooms were standardized across pens and rooms. The north side of each room was equipped with two 91 cm and four 122 cm of diameter wall exhaust fans and two 50.8 cm and 61 cm diameter pit fans. Fig. 1 illustrates the location of rooms in the barn as well as the entry points and place where air samplings took place.

### 2.3. Pig care

All piglets were injected 0.1 ml of Excede (Pfizer Animal Health) at birth. One week after weaning, the following therapy was provided to pigs via water: 32 oz/gal of Vitamin E 40% (400 IU of Vitamin E per gram, AHC Products, Inc.), Ileitis vaccine (Boehringer Ingelheim Vetmedica Inc.) and Erysipela vaccine (Boehringer Ingelheim Vetmedica Inc.). The day before PRRSv experimental infection (−1 dpi), pigs were administered via water 4 oz/gal of ProPrin LQ (12% acetylsalicylic acid, AniMed AHC Products, Inc.) and 0.67 pkg/gal of PenAqua Sol-G (0.5 billion I.U. Penicillin G Potassium, Bimedia, Inc.).

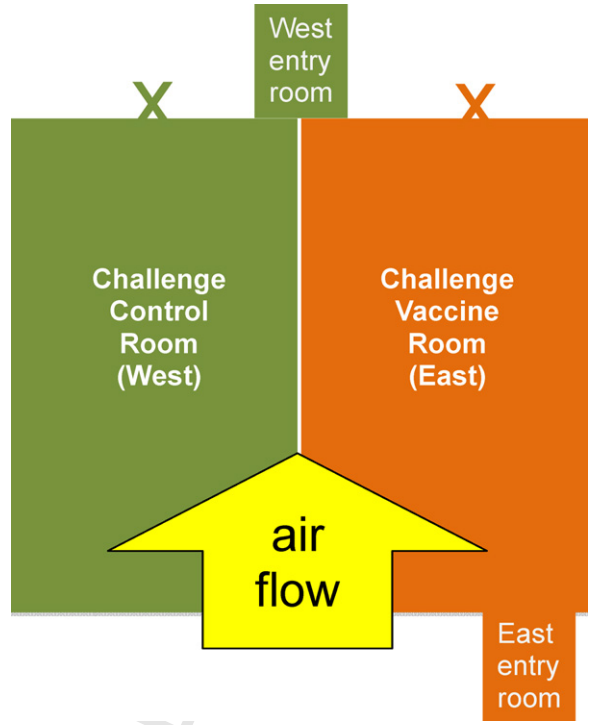


**Fig. 1.** Illustration of the barn, rooms, entry rooms and air flow. Moreover, the "X" show points where air collectors were placed.

### 2.4. PRRS virus isolate

A PRRSv of restriction fragment length polymorphism (RFLP) pattern 1-18-2 belonging to Dr Fred Leung's Lineage 1, sublineage 5 group [25], which is the same lineage as MN 184 type viruses, was isolated from a pig production system located in South Dakota experiencing PRRSv-associated clinical signs. In that system, abortions were at levels below 2.0% and pre-weaning mortality was below 10% for 12 months before the outbreak and the pigs were considered PRRSv-free based on ELISA from sows and RT-PCR from piglets. During winter 2009 sows became off-feed, abortions spiked to 30% and pre-weaning mortality reached 40%. Piglets became PRRSv-RNA positive by PCR and sows seroconverted to ELISA (IDEXX PRRS 2XR Ab Test). Serum sample was drawn from a 20 days-old male piglet with poor hair coat and submitted to the University of Minnesota's Veterinary Diagnostic Laboratory (MN-VDL). After 2 passages in a porcine alveolar macrophage (PAM) cell line, the isolate had a viral concentration of $10^7$ $TCID_{50}$/ml. The inoculum was screened by PCR-based assays at the MN-VDL and tested negative for adventitious agents including porcine circovirus type 1, porcine circovirus type 2, porcine parvovirus, bovine viral diarrhea virus and pseudorabies virus.

### 2.5. Experimental design

Upon arrival, pigs were sorted by gender, source and size and were randomly allocated to the west and east rooms. One thousand and five pigs were housed in the west (hereafter called

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**Table 1**
Pig sample collection and testing time scheme.[a]

| | Days post inoculation | | | | | | |
|---|---|---|---|---|---|---|---|
| | −7 | 0 | 8 | 36 | 70 | 96 | 118 |
| Serum PCR | X | X | X | X | X | X | X |
| Serum ELISA | X | X | X | X | X | X | X |
| Oral fluids PCR | X | X | X | X | X | X | X |
| Tonsil scraping | | | | | X | X | X |

[a] In addition, air samples were collected out of exhaust fans of both groups 6 days a week from day 1 to 118 post experimental inoculation (dpi).

challenge-control room) room and 1009 were housed in the east (challenge-vaccine room) room. At 8 weeks of age, 100 pigs of each room were conveniently selected (2–3 pigs/pen) and intramuscularly inoculated with 2 ml of PRRSv. At 8 and 36 days post infection, all pigs of the challenge-vaccine group were vaccinated with 2 ml of Ingelvac® PRRS ATP (Boehringer Ingelheim Vetmedica Inc.) vaccine. At the same time, pigs of the control room were sham-inoculated with 2 ml of saline solution.

### 2.6. Environmental conditions

The room temperature in the challenge-control group averaged 23.0 °C from 0 to 118 days post experimental inoculation (DPI) (range 18.5–27.2). In the challenge vaccine group, the room temperature averaged 23.1 °C (range 19.1–27.5) in the same period.

### 2.7. Monitoring of PRRSv infection

#### 2.7.1. Animal sampling

The protocol used to monitor PRRSv infection is illustrated in Table 1. Briefly, 35 piglets of each group were randomly selected (no more than 1 pig per pen), identified with ear-tag and designated as "monitor pigs". Blood samples were drawn from monitor pigs into sterile blood vacutainer tubes (Becton-Dickinson Vacutainer, Franklin Lakes, NJ) via jugular venipuncture at −7, 0, 8, 36, 70 96 and 118 days post infection (DPI) and submitted for PRRSv RT-PCR [26] in pools of 5 and PRRSv ELISA tests individually (Herd Chek PRRS Antibody 2XR Test Kit, IDEXX Laboratories, Westbrook, ME). Additionally, 12 pens of each room were randomly identified and had oral fluids collected as described by Prickett et al. [27] at the same points in time and submitted for PRRSv RNA detection by RT-PCR [28]. Also, 12 monitor pigs were randomly selected in each room and had their tonsils scraped at days 70, 96 and 118 dpi (Table 1) using the technique described by Wills et al. [29] and submitted for PRRSv RNA detection using RT-PCR [28].

#### 2.7.2. Air sampling

Two identical liquid cyclonic collectors (Midwest MicroTek, Brookings, SD) capable of collecting 400 l of air per minute were used as described by Dee et al. [30] to collect air coming out of each room (challenge-control and challenge-vaccine) every day from 08:00 AM to 08:30 AM from 0 to 118 days post infection except Sundays. The cyclonic collectors were placed at 1.6 m above the ground, being at the distance of 2 m from and at the same high of the center of the 122 cm of diameter exhaust fan. One cyclonic collector was used for each room. The air samples were kept frozen at −20 °C for up to 10 days and then frozen at −80 °C until tested for PRRSv RNA detection by RT-PCR [30].

#### 2.7.3. Sample testing

The PRRSv RT-PCR and PRRSv ELISA tests from serum samples were performed at the Boehringer Ingelheim Vetmedica Inc. Health Management Center, Ames, Iowa. The RT-PCR for detection of PRRSv RNA from oral fluids and tonsil scrapping samples were performed at the Iowa State University Veterinary Diagnostic Laboratory (ISU VDL), Ames, Iowa. The PRRSv quantitative RT-PCR from air samples were performed at the University of Minnesota Veterinary Diagnostic Laboratory (UMN VDL), Saint Paul, Minnesota. All serum, oral fluids and tonsil scraping samples that were PRRSv RNA positive in the RT-PCR test were submitted for open reading frame 5 (ORF 5) nucleotide sequencing at the ISU VDL. The air samples that were PRRSv RNA positive by RT-PCR were submitted to the UMN VDL for ORF5 nucleotide sequencing.

#### 2.7.4. Biosecurity procedures

Extensive effort was placed on biosecurity to reduce the risk of transferring the vaccine virus from one room to another using scientifically validated protocols of biosecurity [31,32]. The inlets were equipped with insect screening. Each room of the barn had a designated entrance involving a shower-in shower-out facility and room-specific coveralls, boots and equipment. For air collection, there was a dedicated cyclonic collector for each room, which was disinfected everyday before and after use according to protocols previously validated [30].

#### 2.7.5. Growing performance

Frequency of mortality, removal rate and antibiotic treatment were compared between groups. Moreover, feed efficiency and average daily gain were computed from 37 to 137 days of life and compared between groups.

### 2.8. Statistical analysis

The Fisher's exact test was used for between-groups comparison of the difference on the frequency of positive results for serum ELISA, serum PCR, oral fluids PCR and tonsil scrapings PCR. To compare the average of ELISA S/P ratios, the paired *t*-test was used. To compare the frequency of air positive days the McNemar's test was used, and to compare the RNA copies/ml concentration of air positive days and the distribution curve of positive days the Mann–Whitney–Wilcoxon test was used. For the growing performance comparisons, frequency of mortality, removal to sick pens and injectable antibiotic use were compared using Chi-squared test and average daily gain and feed efficiency were compared with paired *t*-test. All the analyses were performed using SAS 9.2 statistical software and the level of statistical significance considered was of $\alpha = 0.05$.

## 3. Results

### 3.1. Establishment of infection

The population of pigs used in this study was confirmed to be serum- and oral fluids-PRRSv RNA negative by RT-PCR assay as well as anti-PRRSv antibody negative at 5 days before challenge. The experimental infection was successfully established as determined by serocoversion of 1/3 of monitor pigs at 8 dpi, detection PRRSv viremia in all monitor pigs at the same point in time and 100% seroconversion of monitor pigs at 36 dpi.

### 3.2. Effect of the vaccine on PRRSv infection dynamics

There was no significant difference between groups in the PRRSv infection dynamics measured as viremia level and duration, antibody response and presence of PRRSv RNA in the tonsils (Table 2).

### 3.3. Effect of the vaccine on viral shedding via oral fluids

PRRSv RNA was detected in oral fluids of both rooms at 8 dpi. At 36 dpi, the proportion of PCR-positive oral fluid samples was

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**Table 2**
PRRSv infection monitoring qualitative results at different points in time shown as number of positive results/total number of tested samples (percentage of positive results).

| Sample | Group[b] | Days post inoculation | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | −7 | 0 | 8 | 36 | 70 | 96 | 118 |
| Serum PCR[a] | Ctrl | 0/7 (0.0%) | 0/7 (0.0%) | 7/7 (100%) | 4/7 (57%) | 0/7 (0.0%) | 0/7 (0.0%) | 0/7 (0.0%) |
| Positive/total (% positive) | Vac | 0/7 (0.0%) | 0/7 (0.0%) | 7/7 (100%) | 4/7 (57%) | 0/7 (0.0%) | 0/7 (0.0%) | 0/7 (0.0%) |
| Serum ELISA | Ctr | 0/35 (0%) | 0/35 (0%) | 11/35 (31%) | 35/35 (100%) | 35/35 (100%) | 33/34 (97%) | 32/34 (94%) |
| Positive/total (% positive) | Vac | 0/35 (0%) | 0/35 (0%) | 12/35 (34%) | 35/35 (100%) | 35/35 (100%) | 34/34 (100%) | 34/34 (100%) |
| Oral fluids PCR | Ctr | 0/7 (0.0%) | 0/7 (0.0%) | 2/6 (33%) | 6/6* (100%) | 0/7 (0.0%) | 0/7 (0.0%) | 0/7 (0.0%) |
| Positive/total (% positive) | Vac | 0/7 (0.0%) | 0/7 (0.0%) | 4/6 (66%) | 2/6 (33%) | 0/7 (0.0%) | 0/7 (0.0%) | 0/7 (0.0%) |
| Tonsil scrapping PCR | Ctrl | – | – | – | – | 6/12 (50%) | 5/12 (42%) | 2/12 (17%) |
| Positive/total (% positive) | Vac | – | – | – | – | 4/12 (33%) | 5/12 (42%) | 1/12 (8%) |

[a] Samples tested for PRRSv RNA by PCR in pools of 5.
[b] Ctrl, challenge-control group; Vac, challenge-vaccine group.
* Statistical significance between groups at alpha level of 0.05.

**Table 3**
Summary of PRRSv RNA detected by qRT-PCR from air samples.

| | Challenge-control group | Challenge-vaccine group |
|---|---|---|
| *PRRSv concentration (RNA copies/ml)* | | |
| Mean | $4.3 \times 10^8$ | $4.5 \times 10^8$ |
| Minimum | $1.2 \times 10^4$ | $4.9 \times 10^3$ |
| Maximum | $1.9 \times 10^9$ | $2 \times 10^9$ |
| 95% conf interval | $(2.6 \times 10^8, 6.1 \times 10^8)$ | $(2.7 \times 10^8, 8.1 \times 10^8)$ |
| *Frequency of air-PRRSv qPCR positive days* | | |
| Number of positive days | 31 | 17* |
| Duration (last positive day) | 70 | 45 |

* Statistical significance between groups at alpha level of 0.05.

significantly less in the challenge-vaccine group ($p = 0.03$). After 36 dpi, all collected oral fluids samples from both groups were PCR-negative.

### 3.4. Effect of the vaccine on viral shedding in bio-aerosols

The summary of PRRSv RNA findings in air samples by qRT-PCR are summarized in Table 3 and illustrated in Figs. 2–4. There was no difference in PRRSv RNA concentration in the air samples between groups. However, the challenge-vaccine group had a significant reduction in the total number of days that PRRSv RNA was detected in air samples (14 days, $p = 0.0004$). Moreover, the challenge-vaccine group had a shorter period of time (25 days) that had air samples positive for PRRSv RNA.

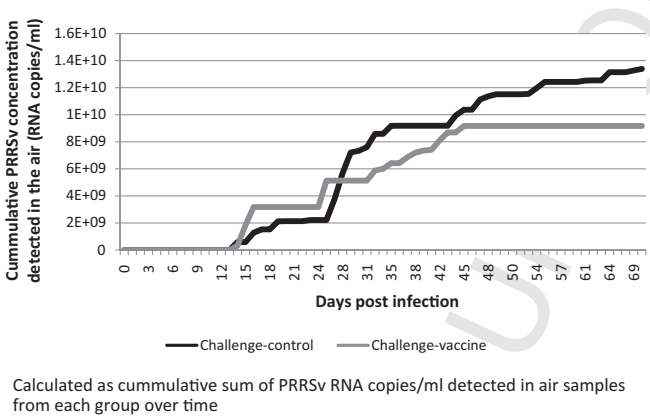


**Fig. 2.** Cumulative days of PRRSv RNA* detected in air samples by RT-PCR. Calculated as cummulative sum of PRRSv RNA copies/ml detected in air samples from each group over time.

### 3.5. PRRSv differentiation

To determine whether the PRRSv RNA detected in each sample was from the (a) PRRSv isolate used for experimental challenge, (b) from the vaccine virus or (c) from new lateral virus introduction, PRRSv PCR-positive samples were submitted for virus differentiation by ORF 5 sequencing. The PRRSv ORF 5 sequence phylogenetic comparison of the sequenced samples is shown in Fig. 5.

In both groups, the PRRSv RNA detected in the air, serum and tonsil scraping samples were at least 99.0% similar to the wild type virus based on ORF 5 nucleotide sequences, with the exception of the air samples detected in the challenge-vaccine group at 15 and 16 dpi (7 and 8 days post first vaccination) that were identical to vaccine strain (Fig. 5). The viruses detected in oral fluids could not be differentiated.

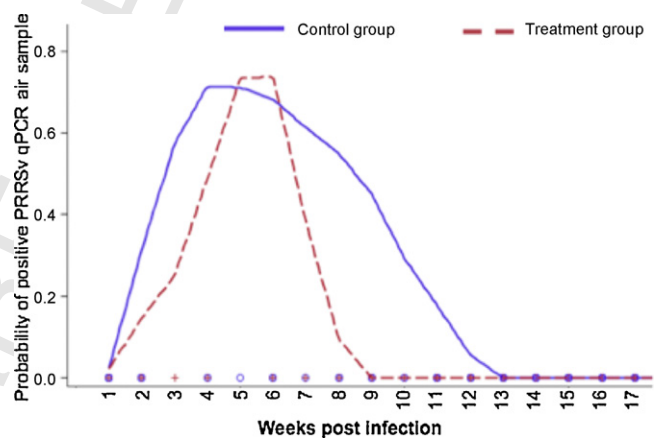


**Fig. 3.** Probability curves of control and treatment group to have air positive days.

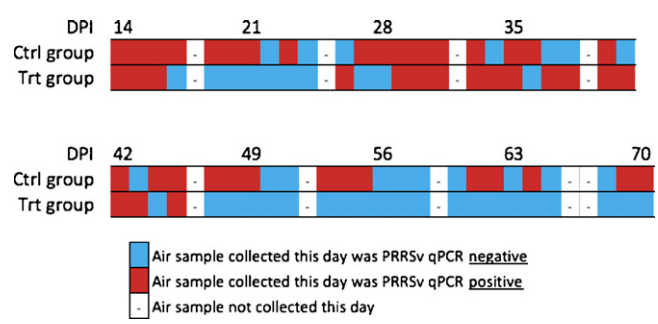


**Fig. 4.** Detailed air sample PRRSv qPCR results from 14 to 70 days post experimental infection. (All samples collected from 0 to 13 and after 70 dpi resulted PRRSv RNA-negative by qRT-PCR.)

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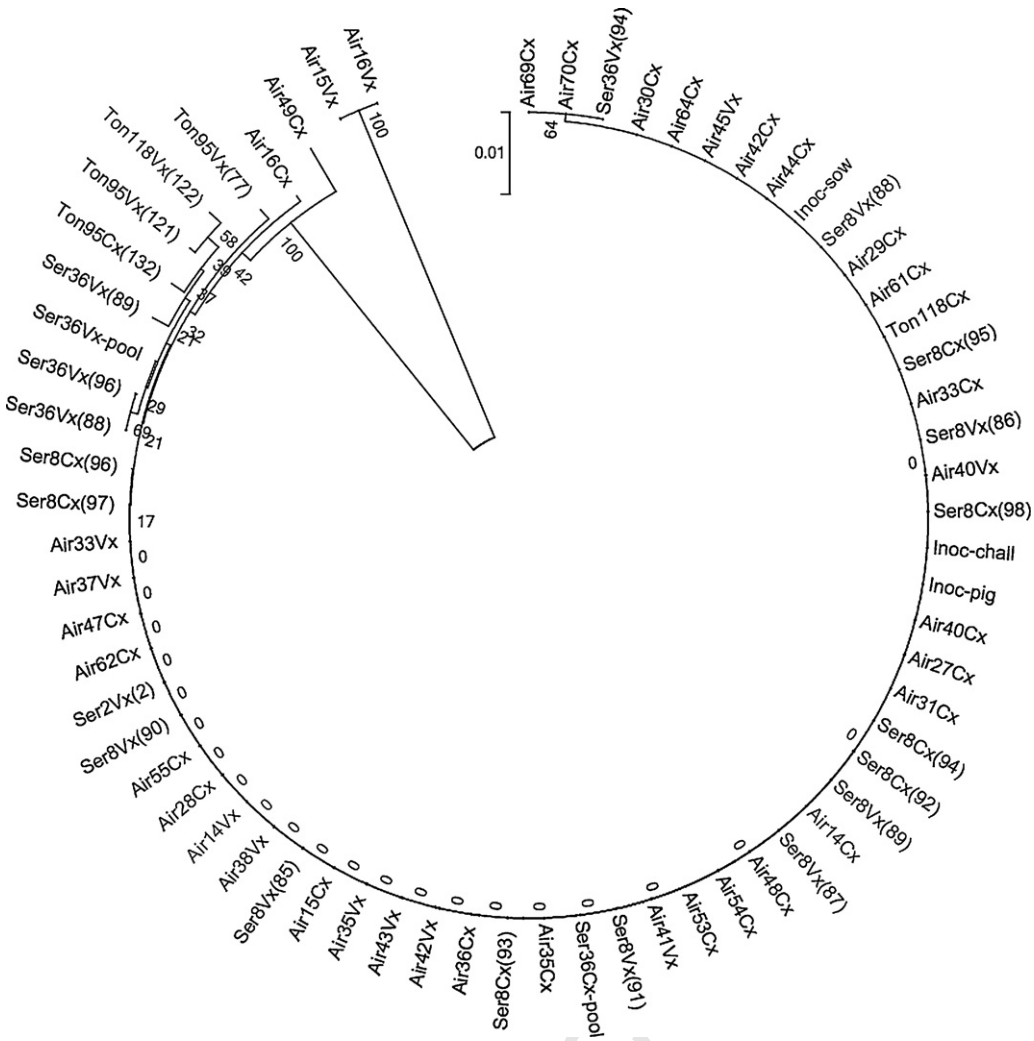


**Fig. 5.** Phylogenetic comparison between PRRSv-ORF 5 nucleotide sequences. (The evolutionary history was inferred using the Neighbor-Joining method [37]. The optimal tree with the sum of branch length = 0.10219240 is shown. The tree is drawn to scale, with branch lengths in the same units as those of the evolutionary distances used to infer the phylogenetic tree. The evolutionary distances were computed using the Maximum Composite Likelihood method [38] and are in the units of the number of base substitutions per site. The analysis involved 66 nucleotide sequences. All positions containing gaps and missing data were eliminated. There were a total of 591 positions in the final dataset. Evolutionary analyses were conducted in MEGA5 [39].)

### 3.6. *Effect of the vaccine on growing performance*

There was no difference in the number of deaths ($p = 0.58$), number of pigs removed to sick pens ($p = 0.33$) and number of pigs treated with injectable antibiotics ($p = 0.98$) between groups. The overall mortality in both groups was 1.5%. The average daily gain (from 37 to 113 days) in the challenge-control and challenge-vaccine group was 1.70 and 1.68 respectively ($p = 0.42$), and the feed conversion rate for the same time period was respectively 2.18 and 2.20 ($p = 0.22$). In summary, despite the evidence of PRRSv infection, there was no apparent PRRS clinical manifestation in the study groups.

## 4. Discussion

Validation of strategies to minimize PRRSv shedding are essential to prevent PRRSv dissemination and to assist in the regional control and elimination of PRRSv. This study evaluated, for the first time, the effectiveness of a PRRS modified-live virus vaccine to reduce viral shedding through oral fluids and bio-aerosols in an infected large growing pig population under field conditions. The study was conducted in a research wean-to-finish barn that resembles typical commercial growing pig farms located in the Mid Western United States.

The PRRSv inoculum used in this study was isolated from a pig production system that attributed extreme piglet mortality and reproduction problems to the strain used (RFLP pattern 1-18-2). Moreover, the veterinarian responsible for that production system reported that this same strain (with similar ORF 5 sequence) was detected in multiple unrelated farms suggesting its potential airborne transmission capacity. PRRSv ORF 5 sequencing (Fig. 5) showed 100% nucleotide similarity between the production system, the inoculum that was prepared for this study and viremic pigs at 2 and 8 days post experimental infection, which suggested that the inoculum used in this study was similar to the one that caused infection and economic impact in the aforementioned pig production system.

To mimic PRRSv infection in commercial pig barns, 10% of piglets were inoculated with a live PRRSv isolate. At least two pigs per pen were infected to assure that all non-infected pigs had a chance to make effective contacts with inoculated pigs and become infected naturally. Regarding the PRRSv infection dynamics observed in the study, the timing of viremia was in line with the expected based on previous reports [12,33], and seroconversion happened sooner than reports from other studies [33–35].

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Considering that PRRSv causes prolonged infection in lymphoid tissues [11], in this study tonsil scrapings were collected from monitor pigs at 70, 96 and 118 dpi to have an assessment of the effect of Ingelvac® PRRS ATP vaccine on the duration of PRRSv infection. There was no difference in the number of PRRSv RNA-positive tonsil scraping samples between groups.

Although the frequency of oral fluid samplings was limited, under the conditions of this study the oral fluids PRRSv RT-PCR results appeared to be correlated with detection of PRRSv qPCR results in air samples. At around 36 dpi the control group had 6 out of 6 oral fluids PCR positive and the air samples were frequently positive around 36 dpi. One month later (at 70 dpi) all oral fluids were negative. After 70 dpi there were no days with air samples PCR-positive. Likewise, in the challenge-vaccine group at 36 dpi 2 out of 6 oral fluid samples were positive and the last PCR-positive air sample in the challenge-vaccine group was soon after that, at 45 dpi. Further studies with higher frequency of oral fluid samplings that also include air samplings are necessary to further investigate this association.

Results showed that although vaccine virus established infection, the wild-type isolate remained the dominant isolate in the challenge-vaccine group population. In other words, the vaccine isolate could not replace wild-type virus under the conditions of this study, which was also reported by others [12,23]. Taking all this information together, we speculate that when PRRSv-positive pig populations are vaccinated with PRRS modified-live virus vaccine, it is expected that the wild-type virus remains dominant and that there is limited regional spread of vaccine virus through bio-aerosols.

The PRRSv ORF-5 sequence phylogenetic comparison of sequences (Fig. 5) illustrates that in this study the PRRSv changed less than 2% across the 118 days post infection, which was the rate of change that was expected. The estimated rate of nucleotide substitution for PRRSv is from 4.7 to $9.8 \times 10^{-2}$ per year [36,37], the highest ever reported among all RNA viruses. We did not investigate the reason why oral fluids samples could not be differentiated.

Under the conditions of this study, vaccinating with PRRS MLV a population of previously infected pigs did not negatively impact growing performance (no difference in mortality, average daily gain and feed efficiency between groups). This finding supports the practice of administering PRRSv MLV vaccine preventively in growing pigs housed in sites with high risk of PRRSv-infection (i.e. high density regions).

Therapeutic vaccination, as described in this study, is another option that can be used to reduce PRRSv regional dissemination from infected pig populations. A possible explanation for the effect of vaccination on viral shedding reduction is that even though PRRSv is highly infectious, it is not highly contagious which means that once PRRSv infects a growing pig population, it takes time to spread in the barn. Conversely, when a live virus is inoculated in all pigs, at once every animal is removed from the "susceptible" pool and is given the chance to develop immunity. In other words, at the population level, the duration of shedding is expected to be less in populations mass-exposed with a live virus because the probability of transmission is reduced as a result of immunity [12].

It is reasonable to speculate that the effect of one dose of attenuated vaccine would be similar to that of two doses. The benefit of two shots is to assure that the right dose was effectively administered in 100% of the pigs. Future studies are needed to further evaluate that.

Regarding type of vaccine, up to this moment, there is no scientific evidence that killed PRRSv vaccine can develop protective immunity or reduce shedding in pigs [38]. Ingelvac® PRRS ATP vaccine was chosen as attenuated vaccine for this study because it was the product being used by the production system that participated in this study. Considering that the degree of genetic homology of PRRSv ORF5 between challenge isolate and attenuated vaccine is not a good predictor of vaccine efficacy [39], we speculate that similar results could be expected if Ingelvac® PRRS MLV (11.3% divergent from Ingelvac® PRRS ATP based on ORF5-sequence) or another attenuated PRRSv vaccine was used.

The strengths of this study included that the experiment was performed under field conditions in a population of 2000 pigs with a PRRSv isolate that was circulating and causing clinical signs associated with PRRS in that region. This study shows for the first time that PRRSv modified-live virus vaccine is an effective tool to reduce the PRRSv load in the air produced by PRRSv infected pig populations. Pigs from the vaccinated group had reduced frequency of air PRRSv positive days ($p = 0.0004$).

This study had some limitations, which include the fact that there was a single air sampling per day that collected about 12,000 l of air coming out of each room (group). Likewise, oral fluids and serum sampling took place only on a monthly basis, which was enough to answer the study's main question but did not allow proper correlation between the outcomes of the different techniques (PCR from serum, oral fluids and air). Multiple air samplings per day and increased frequency of blood and oral fluids collection would give more statistical power and be more representative of the true PRRSv infection and shedding dynamics. Moreover, although there were more than 1000 pigs per group, the study as a whole was a single replicate that evaluated the effect of the treatment (vaccine) against a single PRRSv wild-type isolate and with a single pig genetic line. Therefore, futures studies to address those limitations would be of great value.

## 5. Conclusions

This study reports for the first time the use of PRRSv modified-live virus vaccine as a tool to reduce viral shedding to the environment, measured with PRRSv RT-PCR from oral fluids and bio-aerosols collected with a previously validated air sampler device. Despite the limitations of the study, results support that vaccinating infected growing pig populations with PRRSv MLV vaccine is a potential method to enhance the success of PRRSv regional control because it significantly reduces the cumulative virus shed in the air. Additional studies using different PRRSv strains and different field conditions would be helpful to further understand the findings reported in this study.

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